

2 A student investigated the oxygen consumption of isolated mitochondria under different conditions.

Mitochondria were isolated by crushing liver cells in ice cold buffer and filtering to remove cell debris. The filtrate was centrifuged to separate the mitochondria. The mitochondria were suspended in ice cold buffer.

(a) State **one** reason for keeping the mitochondria in cold buffer solution during isolation.

.....
..... [1]

Fig. 2.1 shows the apparatus used to measure the oxygen consumption of the isolated mitochondria.

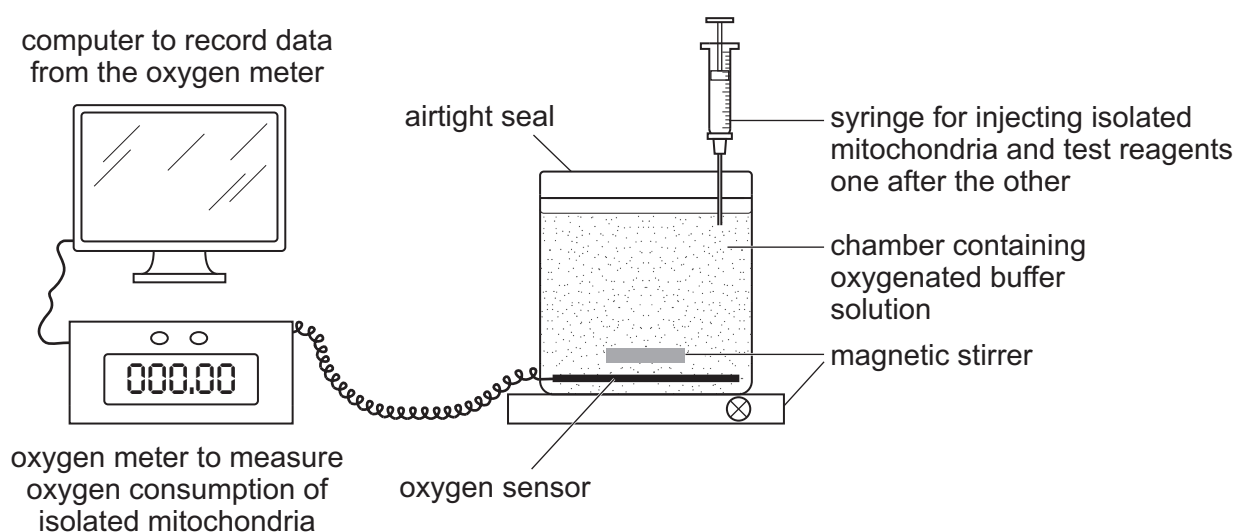


Fig. 2.1

The apparatus was used as follows.

- The apparatus was set up as shown in Fig. 2.1 and left for 1 minute to equilibrate.
- At 1 minute, a standard volume of mitochondria in buffer solution was injected through the seal.
- At 2 minutes, a solution of succinate, a Krebs cycle intermediate, was injected through the seal.
- At 5 minutes, a solution of ADP was injected through the seal.
- At 6 minutes, a solution of cyanide was injected through the seal.

During this procedure the oxygen concentration in the chamber was measured continuously and displayed as a trace on the computer screen as shown in Fig. 2.2.

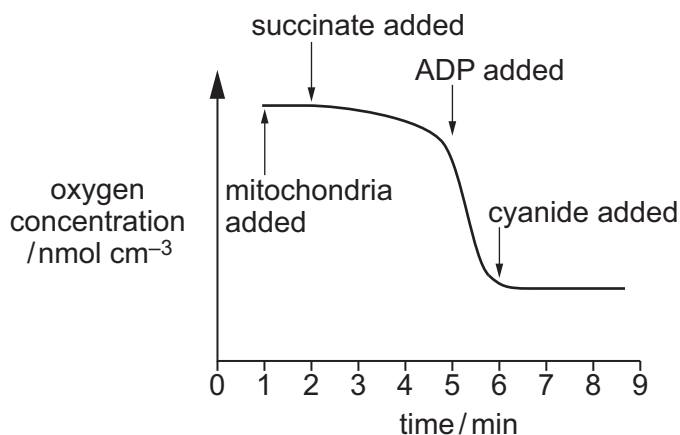


Fig. 2.2

Using the figures from the trace, the computer calculated the rate of oxygen consumption following each addition. This procedure was repeated three more times using fresh samples of mitochondria.

Table 2.1 shows the rates of oxygen consumption for each of the four trials.

	Table 2.1				
	rate of oxygen consumption / nmol min ⁻¹				
	trial 1	trial 2	trial 3	trial 4	mean
mitochondria alone	0.03	0.02	0.03	0.01	0.02
mitochondria with succinate	50.22	49.10	48.53	50.15	
mitochondria with ADP and succinate	139.23	170.10	142.67	138.10	147.53
mitochondria with cyanide, ADP and succinate	0.00	0.10	0.00	0.01	0.03

(b) (i) Calculate the mean rate of oxygen consumption for mitochondria with succinate.

Write your answer in Table 2.1. [1]

(ii) On Table 2.1 indicate, by placing a circle around each value, **two** results that may be anomalous. [2]

(iii) State **two** ways that the student could have processed the results to allow for these possible anomalies.

.....

.....

.....

..... [2]



(c) Explain the results shown in Fig. 2.2 and Table 2.1.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 9]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.